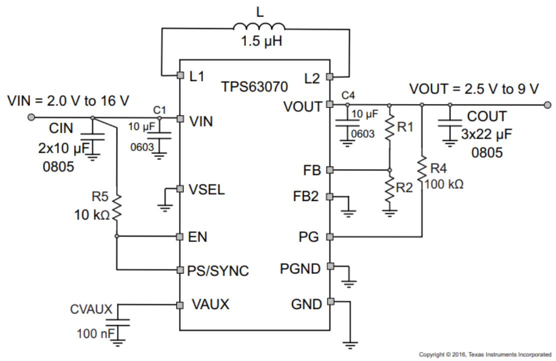
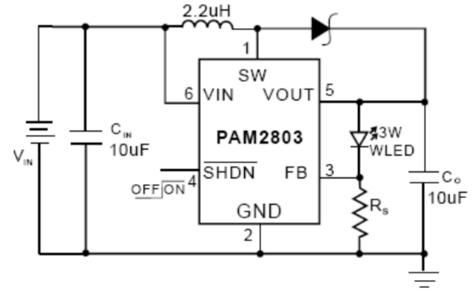


Figure 10. Typical Application For Adjustable Version

LED Current Setting

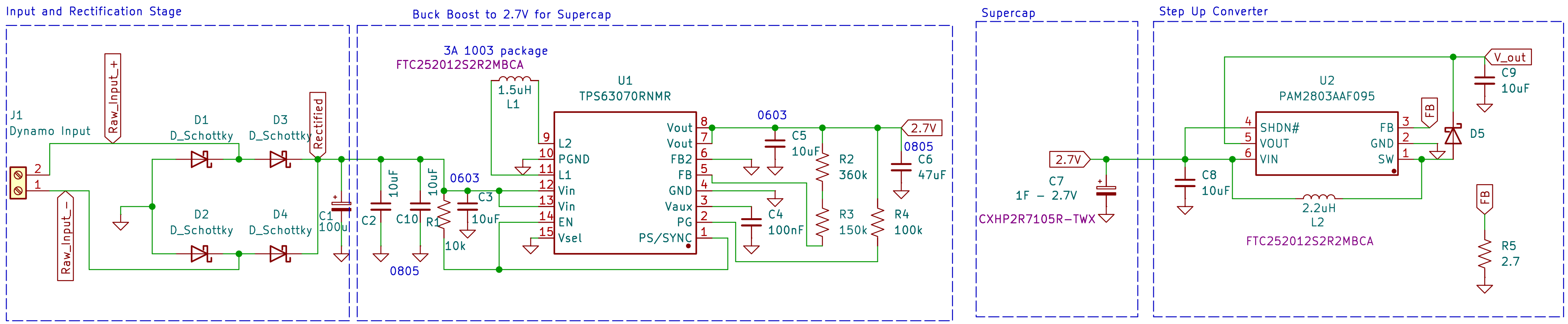
The LED current is set by the single external R_S resistor connected to the FB pin as shown in the typical application circuit on Page 1. The typical FB reference is internally regulated to 95mV. The LED current is $95mV/R_S$. It's recommended to use a 1% or better precision resistor for the better LED current accuracy. The formula and table 1 for R_S selection are shown as follows:

$$R_S = 95mV / I_{LED}$$



I_{LED} = 750mA, R_S = 0.127Ω

Limiting ourselves to 35mA output
with Rs = 2.7 ohms

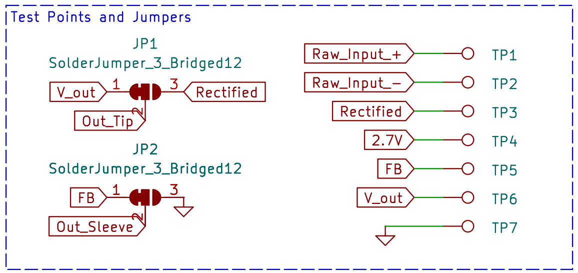


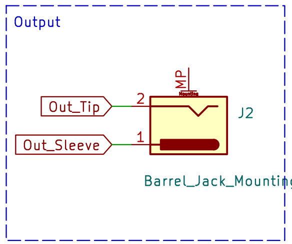
9.2.2.1 Programming The Output Voltage

While the output voltage of the TPS63070 is adjustable, the TPS630701 is set to a fixed voltage. For fixed output versions, the FB pin must be connected to the output directly. The adjustable version can be programmed for output voltages from 2.5V to 9V by using a resistive divider from VOUT to GND. The voltage at the FB pin (V_{REF})is regulated to 800mV. The value of the output voltage is set by the selection of the resistive divider from Equation 6 . It is recommended to choose resistor values which allow a current of at least 2uA, meaning the value of R2 shouldn't exceed 400kΩ. Lower resistor values are recommended for highest accuracy and most robust design.

$$R_1 = R_2\left(\frac{V_{OUT}}{0.8V} - 1\right) \quad (6)$$

R1 = 360k
R2 = 150k
Vout = 2.72V





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